Karageshwaran

Stat 50

Homework #6

(1) $P(A) = 0.8$

$P(A \cap B) = 0.2$

$P(B) = ?$, A and B are independent.

$P(B|A) = P(B)$ when A and B are independent

$P(B) = \frac{P(A \cap B)}{P(A)}$

$P(B) = \frac{0.2}{0.8} = 0.25$

$\boxed{P(B) = 0.25}$

(7)

(a) $P(B) = 0.90$ ⇒ B and N are independent

$P(N) = 0.85$

then, $P(B \cap N) = P(B) \cdot P(N) = 0.90\ (0.85)$

$\boxed{P(B \cap N) = 0.765}$

(b) $P(B^c) = 0.1$

$P(N^c) = 0.15$

then, $P(B^c \cap N^c) = (0.1) \cdot (0.15) = 0.015$

$\boxed{P(B^c \cap N^c) = 0.015}$

(c) $P(B \cup N) = P(B) + P(N) - P(B \cap N)$

$P(B \cup N) = 0.90 + 0.85 - 0.765$

$\boxed{P(B \cup N) = 0.985}$

(15)

(a) $P(A) = 0.90$ A and B are independent

$P(B) = 0.80$

$P(A \cap B) = (0.90)(0.80) = 0.72$

$P(A \cup B) = P(A) + P(B) - P(A \cap B) = 0.9 + 0.8 - 0.72$

$\boxed{P(A \cup B) = 0.98}$

(b) $P(A^c \cup B^c) = 0.1 + 0.2 - 0.28$

$\boxed{P(A^c \cup B^c) = 0.02}$

(c)

$P(A \cap B) = 0.9 (0.8) = 0.72$

$\boxed{P(A \cap B) = 0.72}$

(d) $P(A \cap B^c) = P(A) - P(A \cap B)$

$P(A \cap B^c) = 0.9 - 0.72 = 0.18$

$\boxed{P(A \cap B^c) = 0.18}$

(15)

Lot	conforming	Non-conf.
A	88	12
B	165	35
C	260	40

(a) Lot A:-

$88 + 12 = 100$

$P(\text{conforming "A"}) = 88/100$

$\boxed{P(A) = 0.88}$

(b) conforming :-

$88 + 165 + 260 = 513$

$P(\text{conforming "A"}) = 88/513 = 0.1715$

$\boxed{P(B) = 0.1715}$

(c) Not Lot C :-

$P(\text{not "c"}) = \frac{88 + 165}{513} = \frac{253}{513} = 0.4932$

$\boxed{P(c) = 0.4932}$

(d) Not "c" :-

conformings :- $88 + 165 = 253$

total Not "c" = 300

$P(d) = 253/300 = 0.8433$

$\boxed{P(d) = 0.8433}$

(19)

	Male	Female
Democrat	32	16
Republican	42	8
Independent	2	0

(a) $P(MR) = 42/100 = 0.42 \Rightarrow \boxed{P(MR) = 0.42}$

(b) $P(D \cup F) = P(D) + P(F) - P(D \cap F)$

$P(D \cup F) = 48/100 + 24/100 - 16/100 = 56/100$

$\boxed{P(D \cup F) = 0.56}$

(c) $P(R) = P(MR) + P(FR) = \frac{42}{100} + \frac{8}{100} = \frac{50}{100}$

$\boxed{P(R) = 0.5}$

(d) $P(R^c) = 1 - P(R) = 1 - 0.5 = 0.5$

$\boxed{P(R^c) = 0.5}$

(e) $P(D) = P(MD) + P(FD) = 32/100 + 16/100$

$\boxed{P(D) = \frac{48}{100} = 0.48}$

(f) $P(I) = \frac{2}{100} = 0.02$

$\boxed{P(I) = 0.02}$

(g)

$P(D \cup I) = P(D) + P(I) - P(D \cap I)$

$= 0.48 + 0.02 - 0$

$= 0.50$

$\boxed{P(D \cup I) = 0.50}$

(20)

risks	~~percent~~	Probability of claim
good risks	70%	$P(G) = 0.005$
medium risks	20%	$P(M) = 0.01$
Poor risks	10%	$P(P) = 0.025$

(a) $P(G \cap AG) = P(G) + P(AG)$

~~probability of claim and percentage~~

$P(G \cap AG) = 0.7 (0.005) = 0.705$

$\boxed{P(G \cap AG) = 0.0035}$

(b) $P(A) = P(A_1) + P(A_2) + P(A_3)$

$= P(AG) + P(AM) + P(AP) = 0.005 + 0.01 + 0.025$

$= 0.04$

$\boxed{P(A) = 0.04}$

(a) $P(G) = 0.7$ | $P(B|G) = 0.005$

$P(M) = 0.2$ | $P(B|M) = 0.01$

$P(P) = 0.1$ | $P(B|P) = 0.025$

$$P(G \cap B) = P(B|G)\,P(G) = 0.7\,(0.005)$$

$$\boxed{P(G \cap B) = 0.0035}$$

(b) $$P(B) = P(B|G)\,P(G) + P(B|M)\,P(M) + P(B|P)\,P(P)$$

$$P(B) = 0.005\,(0.7) + 0.01\,(0.2) + 0.025\,(0.1)$$

$$\boxed{P(B) = 0.008}$$

(c) $$P(G|B) = \frac{P(B|G)\,P(G)}{P(B)}$$

$$P(G|B) = \frac{0.005\,(0.7)}{0.008}$$

$$\boxed{P(G|B) = 0.4375}$$

23

(a) $P(A) = 3/10 = 0.3$

$\boxed{P(A) = 0.3}$

(b) $\boxed{P(B|A) = 2/9} = 0.22222$

(c) $P(A \cap B) = P(A) \cdot P(B|A) = 0.3\ (0.22222)$

$\boxed{P(A \cap B) = 0.06667}$

(d) $P(A^c \cap B) = P(A^c) \cdot P(B) = \frac{7}{10} \cdot \frac{3}{9} = 0.2333$

$\boxed{P(A^c \cap B) = 0.2333}$

(e) $P(B) = P(B \cap A) + P(B \cap A^c)$

$P(B) = P(B|A)\,P(A) + P(B|A^c)\,P(A^c) = 2/9 \cdot 3/10 + 3/9 \cdot 7/10 = 3/10$

$\boxed{P(B) = 3/10}$

(f) No, $P(B) \neq P(B|A)$ and $P(A \cap B) \neq P(A) \cdot P(B)$.

24

(a) $P(A) = \frac{300}{1000} = 0.3$

(b) $P(B|A) = \frac{299}{999} = 0.299299299 = 0.2993$

(c) $P(A \cap B) = P(A)\,P(B|A) = 0.3\ (0.2993) = 0.08978979$

$P(A \cap B) = 0.0898$

(d) $P(A^c \cap B) = P(B/A^c)\, P(A^c) = \frac{700}{1000} \cdot \frac{300}{999} = 0.21021021$

$P(A^c \cap B) = 0.2102$

(e) $P(B) = P(B \cap A) + P(B \cap A^c) = P(B/A)P(A) + P(B/A^c)\, P(A^c)$

$= 0.08979 + 0.210210 = 0.3$

(f) $P(A/B) = \frac{P(A \cap B)}{P(B)} = \frac{P(B/A)\, P(A)}{P(B/A)\, P(A) + P(B/A^c) P(A^c)}$

$= \frac{0.0898}{0.3} = 0.2993$

$P(A/B) = 0.2993$

(g) A and B are independent, it is reasonable to treat A and B as though independent. because.

$P(A) \cdot P(B) = P(A \cap B)$ and

$P(B) = P(B/A)$.

(32)

$P(\text{flaw}) = 0.0002$ $\quad P(\text{flaw}^c) = 0.9998$

(a) $P(\text{fail} \mid \text{flaw}) = 0.995$

$P(\text{fail}^c \mid \text{flaw}^c) = 0.99$

$$P(\text{flaw} \mid \text{fail}) = \frac{P(\text{fail} \mid \text{flaw})\, P(\text{flaw})}{P(\text{fail} \mid \text{flaw})P(\text{flaw}) + P(\text{fail} \mid \text{flaw}^c)P(\text{flaw}^c)}$$

$$P(\text{flaw} \mid \text{fail}) = \frac{(0.995)(0.0002)}{(0.995)(0.0002) + (1-0.99)(0.9998)}$$

$$P(\text{flaw} \mid \text{fail}) = 0.0195$$

(b)

(i) Most bottles that fail inspection do not have a flaw.

(c) $$P(\text{flaw}^c \mid \text{fail}^c) = \frac{P(\text{fail}^c \mid \text{flaw}^c)\, P(\text{flaw}^c)}{P(\text{fail}^c \mid \text{flaw}^c)P(\text{flaw}^c) + P(\text{fail}^c \mid \text{flaw})P(\text{flaw})}$$

$$P(\text{flaw}^c \mid \text{fail}^c) = \frac{(0.99)(0.9998)}{(0.9998)(0.99) + (1-0.995)(0.0002)}$$

$$P(\text{flaw}^c \mid \text{fail}^c) = 0.999999$$

(d) (ii) Most bottles that pass inspection do not have a flaw.

(e) Because it says that inspection process is effective in identifying flawed bottles.

(36)

(a) $P(A^c) = 0.05$ $\quad P(A) = 0.95$

$P(B^c) = 0.03$ $\quad P(B) = 0.97$

$P(\text{Sys functions}) = 1 - ((0.05)(0.03)) = 0.9985$

$\boxed{P(\text{Sys function}) = 0.9985}$

(b) Let

$P(\text{Sys function}) = 1 - (P(A^c)\, P(B^c))$

$\boxed{P(A^c) \cdot P(B^c) = 0.9}$ from question.

$\Rightarrow$ $P(\text{Sys function}) = 1 - p^2$

$= 0.9$

$\Rightarrow 1 - 0.9 = p^2 \Rightarrow \boxed{p^2 = 0.1}$

$\boxed{p = 31623}$

(c)

$\Rightarrow 1 - p^3 = 0.90 \Rightarrow p^3 = 1 - 0.9 = 0.1$

$\Rightarrow p^3 = 0.1 \Rightarrow p = \sqrt[3]{0.1}$

$\Rightarrow \boxed{p = 0.4642}$

(3-1)

(a) $P(C^c) = 0.08$ | $P(C) = 0.92$

$P(D^c) = 0.12$ | $P(D) = 0.88$

$P(\text{sys function}) = 1 - (0.08 \times 0.12)$

$= 0.9904$

$\boxed{P(\text{sys function}) = 0.9904}$

(b) Let $P(\text{sys function}) = 0.99 = 1 - p^2$

Then

$\Rightarrow \quad 1 - p^2 = 0.99 \Rightarrow 0.01 = p^2$

$\Rightarrow \quad \sqrt{0.01} = p \Rightarrow \boxed{p = 0.1}$

(c) Let

$P(\text{sys function}) = 0.99 = 1 - p^3$

Then

$\Rightarrow \quad 1 - p^3 = 0.99 \Rightarrow p^3 = 0.01$

$\Rightarrow \quad p = \sqrt[3]{0.1} = 0.2154$

$\boxed{p = 0.2154}$

(d) $P(\text{sys function}) = 0.99 = 1 - (0.5)^x$

$\Rightarrow \quad 1 - 0.99 = (0.5)^x \Rightarrow 0.5 = \sqrt[x]{0.01}$

$\Rightarrow$ When $x = 6$, $\Rightarrow 0.4642$

$x = 7$, $\Rightarrow 0.5179 \Rightarrow$ [x has to 7 or more]